

Appendix B: Methodology and Experiment Protocol

B.1 Purpose

Appendix B records the experiment protocol, including locked parameters, execution steps, retry rules, logging controls, cost controls and documentation-boundary rules.

B.2 Experiment Parameter-Lock Table

Table B.2 records the locked model, API, prompt, retrieval, execution and logging parameters used for the final comparison. Row-specific notes on temperature control and filing-set freezing are provided below the table.

The version identifiers in Table B.2 map to implemented source files in the submitted codebase

(`Dissertation_Submission_Artefacts.zip/04_Codebase/CS958-R0US04-Codebase.zip/`): prompt templates are defined in `app/prompts/prompt_template_registry.py`; the structured-output schema is defined in `app/prompts/output_schema_registry.py`; RAG configuration is defined through `app/config/runtime_settings.py`, `.env.example` and the locked retrieved-context input file `inputs/rag/rag_retrieved_contexts_locked_v1.jsonl`.

Table B.2: Experiment parameter-lock table

Parameter	Locked value	Protocol rationale
Model display name	GPT-5.5 API	Retains the dissertation focus on a next-generation long-context model while holding the model constant across both context-delivery conditions.
Model ID	gpt-5.5	Official model slug; provider-returned model metadata and run timestamp are logged to manage hidden model-update risk.
API endpoint	Responses API	Used for stateless claim-case-condition calls and structured-output execution.
Service tier	Standard	Avoids priority/batch-specific latency or cost behaviour in the core comparison.
Temperature	Not supplied in final API requests	The selected GPT-5.5 Responses API execution path rejected an explicit temperature parameter during runtime validation. Therefore omitted uniformly for both Long_Context and RAG.
Top-p	Not user-configured in the final API requests; provider logs displayed an effective top-p value of 0.98	No explicit top-p parameter was set by the experiment code in the final request path. The provider log displayed top-p as 0.98, which is treated as the provider-side effective/default setting for the executed calls. Comparability is preserved because top-p was not varied by experimental condition and the same request path was applied uniformly across Long_Context and RAG.

Parameter	Locked value	Protocol rationale
Reasoning effort	Medium	Balances classification quality, cost and latency; identical across long-context and RAG-based conditions.
Tools	None	Prevents web search, file search or external tools from contaminating filing-bounded classification.
Structured output mode	JSON schema, strict=true	Forces outputs into the shared schema for parsing, scoring and auditability.
JSON schema version	sec10k_claim_classification_schema_v1.0-final	Shared by both conditions so output structure is not a confound.
Long-context prompt version	LC_PROMPT_v1.0-final	Supplies the full cleaned filing text and the shared classification task.
RAG prompt version	RAG_PROMPT_v1.0-final	Supplies only retrieved chunks from the same cleaned filing corpus and the shared classification task.
RAG config version	RAG_CONFIG_v1.0-final	Locks local vector retrieval, 750-word chunks, 100-word overlap and top-k 8 to prevent retrieval-configuration drift.
Filing set	Latest official SEC Form 10-Ks for Apple, Amazon, JPMorgan Chase, Bank of America, Verizon and Ford, each passing pre-lock smoke test	Provides a smoke-tested complexity gradient across technology, retail/cloud, banking, telecom/infrastructure and automotive/manufacturing disclosures.
Claim-case scale	48 total claim cases; 8 per filing	Large enough for class-specific metrics and qualitative error analysis while remaining manually auditable.
Label-balance target	16 Supported / 16 Contradicted / 16 Insufficient Evidence	Reduces the risk that overall accuracy is dominated by a majority class.
Condition-level calls	96 main calls before any predefined retry: 48 Long_Context calls and 48 RAG calls.	Every claim case is executed once under long-context and once under RAG-based context delivery.
Maximum output tokens	1,500	Keeps structured JSON outputs bounded while allowing evidence references, rationale and missing-context notes.
Execution order	Fixed claim-case order with condition-paired execution; alternate LC/RAG first by case	Reduces time-based provider drift and condition-order bias.
Retry rules	Maximum retry attempts set to 1 for predefined technical failures; all retry attempts logged.	Separates technical failures from model reasoning failures without allowing outcome-driven reruns.

Parameter	Locked value	Protocol rationale
Logging fields	Claim-case, filing, condition, configured and provider-returned model identifiers, prompt/schema/RAG versions, timestamps, tokens, latency, telemetry-estimated cost, retry data, raw/parsed outputs and RAG metadata	Supports reproducibility, cost/latency analysis, qualitative diagnosis and post-run evidence export without implying a separately retained schema-validity field.
Cost ceiling	USD 125 hard ceiling; USD 90 warning threshold retained for monitoring	Allows the agreed 6-filing GPT-5.5 run while preventing uncontrolled expenditure. The hard ceiling is the operational stop control; the warning threshold supports monitoring.
Post-lock calibration	Not allowed except documented technical-blocker amendment before analysis	Prevents prompt, RAG, label, claim-case or scoring changes based on observed results.
RAG retrieval implementation	Preparation-time in-memory cosine-similarity ranking per filing, with persisted chunk and locked context artefacts	Matches the final implementation: contexts were pre-generated and locked before scored execution, then loaded by run_id rather than retrieved dynamically.
Embedding model for RAG index	text-embedding-3-large	Provides a strong fixed retrieval baseline so the RAG condition is not weakened by a low-quality embedding choice.

Temperature note: Because explicit temperature control was not available in the selected GPT-5.5 execution path, exact output determinism cannot be claimed. This limitation was mitigated by applying the same reasoning effort and request path to both conditions, executing the main run within a defined window, preserving prompt hashes, raw outputs and telemetry, and interpreting the findings as a controlled same-model comparison rather than a universal benchmark.

Filing-set note: The six-company filing set was frozen after latest-10-K selection and smoke testing. Accession URLs, report dates, extraction measurements, token estimates and chunk counts are recorded in Appendix C.

B.3 Main Experiment Execution Pipeline

Table B.3 records the predefined sequence from locked inputs to scored outputs. It distinguishes the preparation path, which constructs the overlapping chunk corpus and pre-generates retrieved contexts, from the scored main-execution path, which loads the locked RAG context for each run rather than performing retrieval again.

Table B.3: Main experiment execution pipeline

Step	Protocol action
1	Confirm the locked SEC 10-K corpus, including accession numbers, source URLs, extraction status, section coverage and token-feasibility records.
2	Confirm the locked claim-case manifest, hidden scoring key, filing-reference anchors and exclusion of unresolved ambiguous cases; keep expected labels outside the model-call queue.
3	Confirm the locked model and execution configuration, including the configured model slug, provider-exposed settings, prompt and schema versions, retry rules and cost controls.
4	Assemble Long_Context prompts using the complete cleaned filing and validate input-token feasibility.
5	Load the pre-generated and locked RAG context for each claim-case, produced from the same cleaned filing corpus using the fixed embedding, chunking, overlap and top-k configuration.
6	Execute the controlled API runner across both context-delivery conditions and retain request payloads, raw responses, parsed structured outputs and telemetry.
7	Apply strict JSON-schema enforcement, parse outputs using predefined failure handling, and preserve raw and parsed records.
8	Score predicted labels against the hidden, locked scoring key and generate condition-level scoring outputs.
9	Aggregate classification metrics, token use, telemetry-estimated model-call cost and latency; apply qualitative error coding only after scoring.
10	Interpret the evidence against RQ1–RQ3, bounding conclusions to the selected filings, claim-cases, model, RAG configuration and task.

B.4 Retry and Invalid-Output Rules

Table B.4: Retry and invalid-output rules

Event	Predefined handling	Audit treatment
API timeout / HTTP 429 / HTTP 5xx	Retry the same request once under the predefined technical-failure policy; no prompt, content or context change.	Record retry count, error code and timestamps. If the retry fails, mark technical failure.
Malformed JSON or schema failure	Apply the predefined invalid-output handling without changing the task, evidence or expected label.	If still invalid, record Schema_Failure and preserve the raw output.
Model refusal / unsupported output format	No manual prompt rewriting after lock; treat as invalid output.	Record as technical/qualitative failure, not as a corrected label.
Wrong label, weak rationale or unsupported reasoning	No retry.	Score as submitted model output; code error category where applicable.
RAG retrieval miss discovered after output	No post-hoc retrieval correction.	Code as retrieval miss, noise or fragmentation in qualitative analysis.

Note: In the final main run, no retry was triggered; all 96 condition-level calls completed successfully.

B.5 Logging Fields and Retained Artefacts

Table B.5 records the log and artefact families retained for reproducibility, audit and post-run analysis. The experiment did not rely on a single flat log file; execution metadata, prompt payloads, raw and parsed outputs, telemetry, scoring outputs and RAG-context records were retained separately so hidden expected labels were introduced only after model execution.

Table B.5: Logging fields and retained artefacts

Retained artefact	Main retained fields / content	Purpose
Locked model-call queue	run_id, case_id, filing metadata, condition, prompt/schema/RAG version identifiers and execution order; expected labels excluded.	Defines the model-facing execution queue without label leakage.

Retained artefact	Main retained fields / content	Purpose
Prompt payload records	Model-facing inputs, supplied context, prompt version, schema version and prompt hashes.	Preserves the exact inputs sent to each condition.
Raw response records	Provider response identifier, model identifier, usage details, raw output and provider-displayed execution settings where returned.	Preserves the unprocessed model-call evidence.
Parsed output records	Structured prediction, evidence references, reasoning summary and confidence field.	Supports validation, parsing and later scoring.
Telemetry records	Execution status, elapsed time, input tokens, output tokens, telemetry-estimated cost, retry attempts used and error message where applicable.	Supports cost, latency and execution-status analysis.
Scoring outputs	Predicted label matched against the locked expected label after execution.	Produces condition-level correctness and metric inputs without exposing labels during model calls.
RAG-context records	Retrieved chunk identifiers, ranks, similarity scores, word offsets, chunk text, top-k value and embedding model.	Supports RAG audit and qualitative diagnosis of retrieval miss, noise or fragmentation.

B.6 Official Model-Provider Documentation Boundary

Official SEC and API-provider documentation is used only for official facts, such as filing structure, model/API capability, structured-output behaviour, rate limits and pricing. It is not used as evidence for model or RAG performance claims.